

Deployment

Flask deployment is the process of moving your web application from your local computer (`localhost:5000`) to a live web server so that it is accessible to users worldwide via a public URL. When you run `app.py` locally, you use Flask's built-in **development server**. While great for testing and debugging, this server is single-threaded, insecure, and crashes easily under heavy traffic. Production environments require a robust setup capable of handling concurrent requests safely and efficiently.

Development and Production

In the **development environment**, the programmer tests and modifies the application. Flask's development server and debug mode are useful here.

In the **production environment**, the application is made available to actual users. Security, performance, reliability, and scalability become important.

Flask Deployment Process

The general deployment process consists of:

- Develop the Flask application

Create the application using Python, Flask, HTML, CSS, JavaScript, etc.

- Test the application

Check that the routes, forms, database operations, and other functions work correctly.

- Prepare dependencies

All required Python packages are listed so they can be installed on the production server.

- Configure the application

Production configuration, database settings, secret keys, and environment variables are properly configured.

- Use a production WSGI server

A WSGI server such as Gunicorn receives web requests and passes them to the Flask application.

- Configure a web server

A server such as Nginx can act as a reverse proxy and handle tasks such as incoming connections and static files

- Enable security
HTTPS, secure passwords, authentication, input validation, and appropriate access controls should be configured.
- Monitor and maintain the application
Logs, errors, performance, dependencies, and security should be monitored regularly.

Why is Flask Deployment Important?

- Makes the application accessible to users.
- Provides better performance than the development server.
- Improves security.
- Supports multiple users and requests.
- Allows the application to run continuously on a server.
- Makes it possible to host the application on cloud or dedicated servers.